Technical Appendix: When Can Online Narratives Test Organizational Theory? Evidence, Silence, and LLM Choice

How to Read This Appendix

Section 3 of the proposal summarizes our workflow in four steps. This appendix explains each step with sufficient detail for readers to evaluate and reproduce it. Section A describes the data preparation, measures for focal constructs, and LLMs. Section B explains the four-step workflow, one subsection per step; each begins with a shaded summary of what the step does and why, followed by the procedure. Section C reports a robustness check and planned analyses, Section D documents reproducibility, and Section E gives the full prompts. Table 1 defines the key terms used throughout this appendix.

Table 1

Key Terms

Term	Meaning in this study
LLM	Large language model
Statement (semantic cluster)	A group of proposition occurrences in a post that express the same substantive claim. Each cluster is counted once as a unique idea, while repeated occurrences are retained when calculating RID.
Evidence question	One of five questions describing a focal construct (e.g., "How does the user describe their job satisfaction?"). Used in Step 1 to decide whether a statement is relevant. Not the same as a questionnaire item.
Questionnaire item	One of 20 established questionnaire items rated in Step 2 (e.g., "All in all, satisfied with the job").
Relevant information density (RID)	The share of a post's statements that directly answer at least one evidence question. Low when a post is repetitive or off topic.
Construct coverage breadth (CCB)	The share of the five focal constructs for which a post provides direct evidence. Low when a post addresses only one or two topics.
Evidence requirement (threshold)	A minimum value of RID or CCB that a post must exceed to enter the analysis.
Cell	One combination of a minimum RID and a minimum CCB (19 × 19 = 361 cells).
Estimable cell	A cell in which enough posts remain for the statistical model to produce finite estimates.
Mean imputation; MICE-PMM	Two ways of filling in missing item ratings: with the item's average, or by predicting each missing rating from the post's other ratings (multiple imputation; Section B3).
Latent structural equation modeling (SEM); Path analysis A; Path analysis B	Three statistical models used to estimate the same relationships (Table 9).
Similarity (r^*)	How closely the pattern of the estimated relationships matches the pattern expected from the meta-analysis (−1 to +1).
Agreement with human ratings (V_h)	How closely an LLM's construct scores match the average ratings of human raters on the same posts.

A. Data Preparation, Measures for Focal Constructs, and LLMs

A1. Data and Ethics

Using a Python crawler (PRAW), we collected Reddit posts published between May 6, 2010, and February 24, 2026, in 1,632 subreddits, using 45 remote work search terms such as *telework*, *work from home*, and *hybrid work*. After we removed empty and duplicate posts, 5,715 of 7,448 records remained. The post is the unit of analysis. The study received the first author's institutional ethics approval. Collection strictly followed Reddit's terms of use. We store no usernames, quote no posts verbatim, and release a synthetic dataset with the same structure in place of raw posts.

A2. Theoretical Model and Measures

As a test case, we used one part of the theoretical model from Gajendran et al.'s (2024) meta-analysis: remote work intensity (R) increases autonomy (A), which in turn lowers work-related stress (Y_1) and raises job satisfaction (Y_2) and leader–member exchange (LMX; Y_3). Remote work intensity may also relate to the three outcomes directly. Table 2 lists the measures; full item wording is given in prompt E2 (Section E).

Table 2

Constructs and Items

Construct	Symbol	Items	Scale	Example item	Source
Remote work intensity	R	1	0–7	0 = fully on-site; 7 = fully remote	Study operationalization
Autonomy	A	6	1–7	Discretion in deciding where to work	Gajendran et al. (2015)
Work-related stress	Y_1	3	1–7	Workload feels heavy	Ostroff and Kozlowski (1992)
Job satisfaction	Y_2	3	1–7	All in all, satisfied with the job	Golden and Veiga (2005)
LMX	Y_3	7	1–7	I have a high-quality working relationship with my leader	Graen and Uhl-Bien (1995)

Note. Higher scores indicate higher levels of each construct (e.g., more autonomy, higher work-related stress, greater job satisfaction). The second job-satisfaction item is reverse-worded in the original scale and was reverse-coded for analysis so that higher values indicate greater satisfaction.

A3. LLMs

Twelve LLMs from six developers processed every post with identical task definitions and prompts (Table 3). They span open-weight models, which anyone can download and run locally, and proprietary models, which are accessible only through an application programming interface (API). The released workflow runs on open-weight models, allowing others to reproduce it without paid access.

Table 3

LLMs and Access

Model ID	LLM	Developer	Weights	Total parameters	Model identifier	Access
1	GPT-5	OpenAI	Proprietary	600B† [300–1,000B]	gpt-5	API
2	GPT-5-mini	OpenAI	Proprietary	80B† [30–150B]	gpt-5-mini	API

Model ID	LLM	Developer	Weights	Total parameters	Model identifier	Access
3	Qwen3.8-Flash	Alibaba	Proprietary	125B (6B active)	qwen3.8-flash	API
4	Qwen-Plus	Alibaba	Proprietary	235B† [100–300B]	qwen-plus-latest	API
5	Qwen-Max	Alibaba	Proprietary	235B† [100–300B]	qwen-max	API
6	Qwen3-32B	Alibaba	Open-weight	32.8B	qwen3-32b	Local
7	Llama-3.3-70B	Meta	Open-weight	70B	llama-3.3-70b-instruct	Local
8	Phi-4	Microsoft	Open-weight	14B	phi-4	Local
9	DeepSeek-V4-Pro	DeepSeek	Proprietary	1,000B† [600–1,500B]	deepseek-v4-pro	API
10	DeepSeek-V4.1-Flash	DeepSeek	Proprietary	685B† [600–1,000B]	deepseek-v4.1-flash	API
11	DeepSeek-R1-Qwen3-8B	DeepSeek	Open-weight	8B	deepseek-r1-0528-qwen3-8b	Local
12	Granite-3.3-8B	IBM	Open-weight	8B	granite-3.3-8b-instruct	Local

Note. B = billion. Values are total parameters as reported by the developer, except those marked †. †The developer does not disclose the parameter count; the value shown is our estimate based on public information about the model family and architecture, with a plausible range in brackets. These estimates are used only in the model-size analysis (Table 13), which uses log10-transformed values. Qwen3.8-Flash activates 6B of its 125B parameters for each response.

B. The Four-Step Workflow

Table 4 summarizes the four steps; Sections B1–B4 describe each step in turn.

Table 4

Overview of the Four-Step Workflow

Step	What happens	Performed by	Output	Informs
1. Locating evidence and calculating RID and CCB (B1)	Each LLM splits a post into single statements and identifies which of the five evidence questions each statement directly answers.	LLM (prompt E1)	RID and CCB for every post	RQ1
2. Rating the items (B2)	Each LLM rates the 20 questionnaire items of the five constructs, leaving an item missing when the post gives no evidence for it. Twenty-three postgraduate students in organizational behavior and management each rated 32 posts on the 20 items.	LLM (prompt E2)	Item ratings, with missing values	RQ1, RQ2
3. Estimating relationships (B3)	The theoretical model is estimated under every combination of evidence requirements, missing-data methods, and statistical models.	Statistical software	Seven path coefficients per analysis (25,992 analyses)	RQ1, RQ2
4. Evaluating each LLM (B4)	Estimated relationships are compared with meta-analytic expectations, and ratings with human ratings.	Statistical software	r^*, V_h, α, CFA fit, stability	RQ1, RQ3

Note. Section numbers in parentheses refer to this appendix. RID = relevant information density; CCB = construct coverage breadth; RQ = research question; CFA = confirmatory factor analysis.

B1. Step 1: Locating Evidence and Calculating RID and CCB

> ***What this step does.*** Before any construct is rated, each LLM determines how much usable evidence a post contains. The result is two numbers per post: RID, which captures the share of what a text says that is relevant to the constructs being measured, and CCB, which captures how many of the predefined constructs a text covers. Step 3 uses them to decide which posts enter the analysis.

Procedure

Locating Evidence. Each LLM applied the evidence prompt (E1) to every post and completed three tasks:

1. **Split the post into single statements.** Each statement expresses one claim (e.g., "I work from home three days a week"). Statements that repeat the same meaning are grouped into one *semantic cluster*, and the number of repetitions is recorded.
2. **Mark each statement as substantive or not.** Only greetings, boilerplate, and content-free filler are non-substantive. Facts, experiences, evaluations, emotions, intentions, and concrete questions are substantive, even if they are unrelated to work.
3. **Link each substantive statement to the evidence questions (relating to the constructs) it directly answers.** For the five-construct analysis, we retained direct links to the five focal evidence questions listed in Table 5. A statement counts as evidence only if, on its own, it answers the question. Sharing a keyword, topic, or person with the question is not enough, nor is describing a possible cause or consequence of the topic.

Table 5

The Five Construct-Aligned Evidence Questions

No.	Topic	Evidence question	Focal construct
Q1	Remote work intensity and use	How does the user describe their remote work intensity and whether they are a remote work user or non-user? Does the user report specific frequency or amount of remote work (e.g., days per week, full-time remote)?	Remote work intensity (R)
Q2	Autonomy	How does the user describe their perceived job autonomy, flexibility, or control over work location, schedule, or tasks?	Autonomy (A)
Q3	Job satisfaction	How does the user describe their job satisfaction or dissatisfaction?	Job satisfaction (Y_2)
Q4	Work-related stress	How does the user describe work-related stress, pressure, or strain?	Work-related stress (Y_1)
Q5	Leader-member exchange (LMX)	How does the user describe their relationship with their leader or supervisor?	LMX (Y_3)

Note. The five questions correspond directly to the five constructs in the theoretical model we aim to test. The evidence questions are derived from the definitions of the focal constructs.

Calculating RID and CCB. From each LLM's output, we computed three counts for every post: N_{occ}, the total number of statements, counting each repetition separately; N_{rel}, the number of distinct substantive clusters that directly answer at least one evidence question; and Q_{cov}, the number of evidence questions answered by at least one such cluster. Then:

$$\text{RID} = N_{rel} / N_{occ} \qquad \text{(Equation 1)}$$

$$CCB = Q_{cov} / 5 \qquad \text{(Equation 2)}$$

RID is high when a large share of the post's expressed content directly informs at least one evidence question and low when the post contains repetition or unrelated content. CCB is high when the post supplies evidence for most of the five constructs and low when its evidence is concentrated in one or two. Both measures range from 0 to 1.

B2. Step 2: Rating the Items

> ***What this step does.*** Each LLM reads the posts and rates the 20 questionnaire items of the five constructs listed in Table 2, as a human rater would. The key rule is that an item is rated only when the post provides evidence for it; otherwise, it is left missing rather than guessed.

Procedure

1. **Identify the work situation to rate.** Some posts describe more than one work situation, for example before and after a return-to-office mandate. The LLM first separated these situations and selected one for analysis: the situation described most directly, usually the current one, that provides evidence on the largest number of the five constructs. This prevents ratings from mixing different situations.
2. **Rate each item under the selected situation.** Remote work intensity was rated from 0 (fully on-site) to 7 (fully remote) and the 19 other items from 1 to 7, using only information stated in or directly supported by the post. The LLM was instructed not to infer ratings from stereotypes.
3. **Leave unsupported items missing.** When the post gave no evidence for an item, the LLM returned a missing value (null).

Each item therefore received one of three codes: an *observed rating*; *no textual evidence* (missing); or *no usable output*, when the model failed to return a valid response. Table 6 reports, for each LLM, the median RID and CCB and the share of items left missing.

Checking and Storing Responses

Every response was checked automatically before entering the analysis. A valid response contained exactly one selected work situation, a remote work intensity value from 0 to 7 or missing, and item lists of the correct length (6, 3, 3, and 7 items for autonomy, work-related stress, job satisfaction, and LMX) containing integers from 1 to 7 or missing values. Invalid responses were resubmitted. Each valid response was stored with the model name, post identifier, selected situation, item ratings, and the raw model response. Identical posts were rated once and linked to all of their identifiers.

Table 6

Evidence and Missing Ratings by LLM

Model ID	LLM (developer)	Median RID	Median CCB	Missing: Remote work intensity	Missing: Autonomy	Missing: Work-related stress	Missing: Job satisfaction	Missing: LMX
1	GPT-5 (OpenAI)	.000	.000	.556	.856	.906	.849	.953
2	GPT-5-mini (OpenAI)	.042	.200	.358	.596	.669	.671	.841
3	Qwen3.8-Flash (Alibaba)	.067	.200	.559	.929	.849	.828	.944
4	Qwen-Plus (Alibaba)	.154	.200	.412	.765	.733	.794	.931

Model ID	LLM (developer)	Median RID	Median CCB	Missing: Remote work intensity	Missing: Autonomy	Missing: Work-related stress	Missing: Job satisfaction	Missing: LMX
5	Qwen-Max (Alibaba)	.059	.200	.455	.812	.667	.700	.898
6	Qwen3-32B (Alibaba)	.071	.200	.234	.757	.583	.609	.861
7	Llama-3.3-70B (Meta)	.333	.400	.403	.503	.531	.498	.673
8	Phi-4 (Microsoft)	.000	.000	.233	.517	.487	.501	.713
9	DeepSeek-V4-Pro (DeepSeek)	.011	.200	.517	.904	.864	.832	.943
10	DeepSeek-V4.1-Flash (DeepSeek)	.000	.000	.606	.992	.968	.970	.991
11	DeepSeek-R1-Qwen3-8B (DeepSeek)	.000	.000	.121	.610	.611	.665	.735
12	Granite-3.3-8B (IBM)	.000	.000	.144	.699	.677	.633	.653
-	Median across LLMs	.027	.200	.408	.761	.673	.686	.879

Note. RID and CCB are medians across posts (Equations 1–2). CCB values of .200, .400, .600, .800, and 1.000 correspond to direct evidence for 1, 2, 3, 4, and 5 of the five constructs. A median of .000 means that more than half of the posts contained no statement answering any evidence question, according to that LLM. Missing entries are the proportions of item ratings left missing before imputation (e.g., .556 = 55.6%).

B3. Step 3: Estimating Relationships Under Different Analytic Choices

> ***What this step does.*** Researchers using LLM-derived ratings must make three choices: which posts to include (the minimum RID and CCB), how to fill in missing ratings, and which statistical model to use. Each choice can change the results. Instead of picking one combination, we estimated the same theoretical model under all of them: 361 combinations of minimum RID and CCB, two imputation methods, and three statistical models, or 2,166 analyses per LLM. A conclusion that holds across these analyses does not depend on any particular choice; a conclusion that changes reveals which choice it depends on. This approach is known as a multiverse analysis (Steegen et al. 2016).

The Theoretical Model

All analyses estimate the same seven paths (Table 7):

$$A = a\,R + e_{\mathrm{A}} \qquad \text{(Equation 3)}$$

$$Y_{\mathrm{j}} = c'_{\mathrm{j}}\,R + b_{\mathrm{j}}\,A + e_{\mathrm{j}}, \quad j = 1 \text{ (work-related stress), } 2 \text{ (job satisfaction), } 3 \text{ (LMX)} \qquad \text{(Equation 4)}$$

Table 7

The Seven Paths

Path	From → to	What it captures
a	Remote work intensity → autonomy	Whether more remote work goes with more autonomy
b_1, b_2, b_3	Autonomy → work-related stress, job satisfaction, LMX	The autonomy pathway, holding remote work intensity constant
c'_1, c'_2, c'_3	Remote work intensity → work-related stress, job satisfaction, LMX	The remaining direct association of remote work intensity with each outcome, holding autonomy constant

Choice 1: Evidence Requirements (361 Cells)

For each LLM, we used that LLM's own RID and CCB values. The minimum RID and the minimum CCB each ranged from 0 to .90 in steps of .05, giving 19 × 19 = 361 combinations (cells). A post entered the analysis in a cell when its RID and its CCB both exceeded the minimums. Because LLMs judged evidence differently, the posts retained in the same cell differ across LLMs.

Choice 2: Missing-Data Treatment

Within the posts retained in each cell, missing item ratings were filled in with one of two methods:

- ***Mean imputation*** (baseline) replaced each missing rating with that item's average among the retained posts.
- ***Multiple imputation with predictive mean matching*** (MICE-PMM; van Buuren and Groothuis-Oudshoorn 2011) predicted each missing rating from the post's other item ratings, and then filled it with an actually observed rating from one of the five posts whose predictions were closest. This was repeated to create five completed datasets, and the results were combined using Rubin's (1987) rules (Table 8; Figure 1).

Table 8

Mean Imputation Versus MICE-PMM

Statistical model	*r* between LLM averages	*r** (mean imputation)	*r** (MICE-PMM)	Difference (*SD*)	LLMs with higher *r** under MICE-PMM
Latent SEM	.905	.427	.494	+.067 (.148)	9 of 12
Path analysis A	.989	.581	.577	−.004 (.064)	6 of 12
Path analysis B	.994	.699	.667	−.032 (.046)	4 of 12

Note. Computed on paired cells retaining at least 30 posts and all seven path estimates under both methods, averaged within each LLM and then across LLMs with equal weights. Difference = MICE-PMM − mean imputation; *SD* is across LLMs. All 12 LLMs contributed paired cells in each analysis.

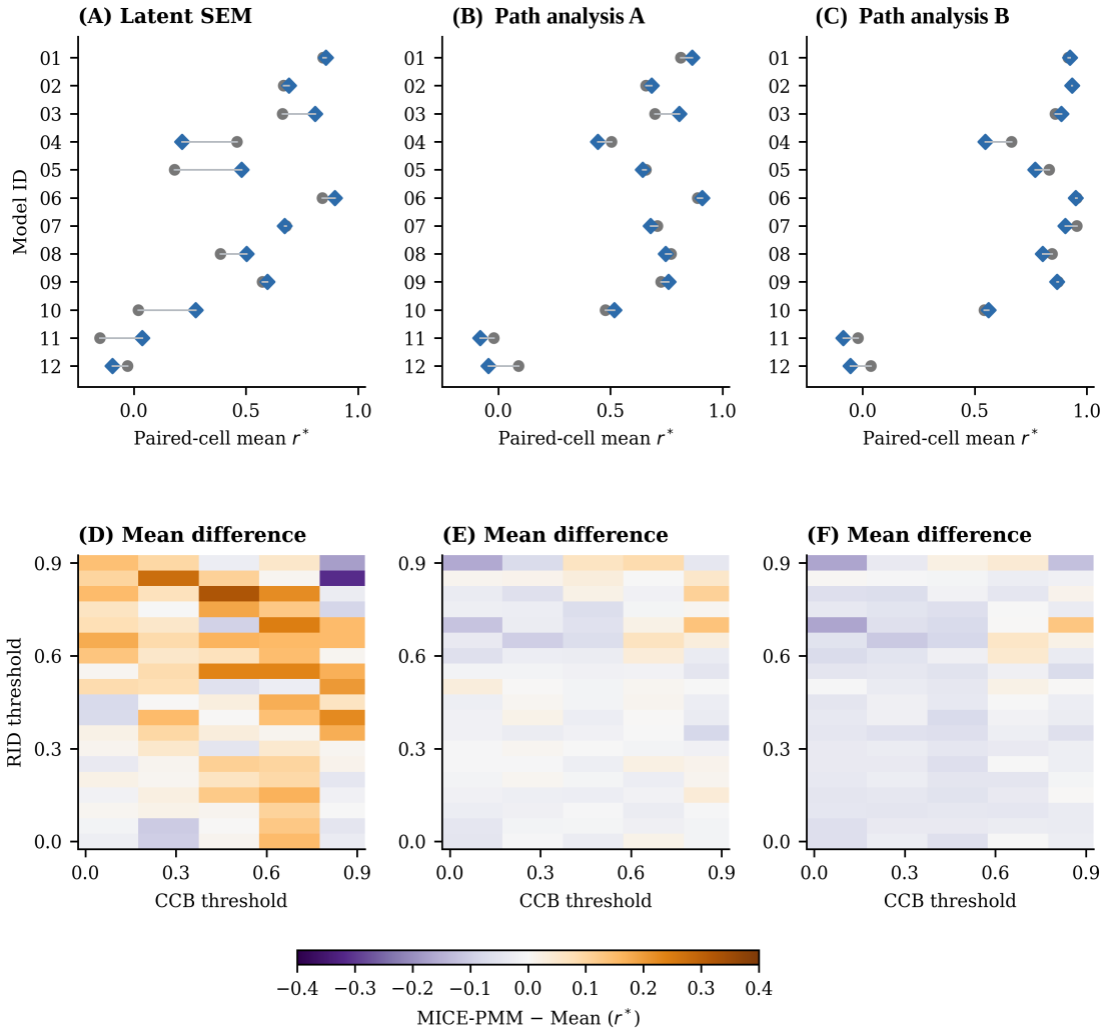


Figure 1. Comparison of mean imputation and MICE-PMM. (A–C) Average *r** for each LLM under mean imputation (gray circles) and MICE-PMM (blue diamonds) for latent SEM, Path analysis A, and Path analysis B; the line between symbols shows the difference. Averages include only threshold combinations estimable under both methods. (D–F) Difference in *r** (MICE-PMM minus mean imputation) at each combination of minimum RID (vertical axis) and minimum CCB (horizontal axis), averaged across LLMs. Orange indicates higher *r** under MICE-PMM, purple higher *r** under mean imputation, and white little difference.

Choice 3: Statistical Model

Each completed dataset was analyzed with three statistical models that differ in how they treat measurement error and the other variables (Table 9). Path analysis A and Path analysis B used standardized scale scores (the mean of each construct's items); all models report standardized coefficients.

Table 9

Three Statistical Models

Model	How constructs enter	What *c′* and *b* estimate
Latent SEM	Autonomy, work-related stress, job satisfaction, and LMX as latent variables	Equations 3–4, corrected for measurement error

Model	How constructs enter	What c' and b estimate
	measured by their items; remote work intensity observed	
Path analysis A (Adjusted composite-score path analysis)	Scale means	Equations 3–4: each path holding the other predictor constant
Path analysis B (Pairwise composite-score path analysis)	Scale means	Seven separate one-predictor regressions: associations without adjustment

Note. Latent SEM was fitted in lavaan (Rosseel 2012) with robust maximum likelihood (MLR). For every analysis, we stored coefficients, standard errors, 95% confidence intervals, sample size, convergence status, and fit indices; for latent SEM, these included CFI, TLI, RMSEA, SRMR, AIC, BIC.

A cell was *estimable* when the model produced finite estimates. In total, the design comprised 12 LLMs × 361 cells × 2 missing-data methods × 3 statistical models = 25,992 analyses.

B4. Step 4: Evaluating Each LLM

> ***What this step does.*** We evaluated each LLM at two levels. At the *relationship level*, we asked whether the relationships estimated from its scores are similar to what the meta-analysis leads us to expect (similarity/concordance, *r**). At the *construct level*, we asked whether its scores match human ratings of the same posts (agreement with human ratings, V_h) and whether its items form coherent scales (α and CFA).

B4.1 Similarity Between the Estimated and Expected Models (r)*

We used *r** to measure the similarity between the estimated model and the model expected from the meta-analysis. Based on Gajendran et al. (2024), we represented the expected model as seven paths, coding each path as +1 when a positive relationship was expected and −1 when a negative relationship was expected (Table 10). For each estimable cell, *r** was calculated as the Pearson correlation between this expected pattern and the seven standardized path coefficients from the estimated model. Higher *r** values indicate greater similarity between the estimated and expected models.

Table 10

Expected Direction of Each Path from Gajendran et al. (2024)

Path	Expected sign
a: Remote work intensity → autonomy	+
b_1: autonomy → work-related stress	−
b_2: autonomy → job satisfaction	+
b_3: autonomy → leader-member exchange (LMX)	+
c'_1: Remote work intensity → work-related stress	−
c'_2: Remote work intensity → job satisfaction	+
c'_3: Remote work intensity → leader-member exchange (LMX)	+

Note. Expected signs follow the focal model in Gajendran et al. (2024).

*Interpretation. r** ranges from −1 to +1. Values near +1 mean that the paths expected to be positive are clearly larger than those expected to be negative; negative values mean a reversed pattern; values near 0

mean no correspondence. For example, hypothetical estimates of (a, c'_1, c'_2, c'_3, b_1, b_2, b_3) = (.40, −.10, .05, .02, −.20, .35, .25) would yield r^* = .78.

Because r^* compares the relative size of the coefficients, it can be high even when some signs are wrong: if every path is positive but the two expected-negative paths are the smallest, as in (.40, .05, .15, .10, .02, .35, .25), r^* is still .71. We therefore also report, for each path, the percentage of estimable cells in which the estimated path had the expected direction (Figure 2).

Aggregation. r^* was averaged across estimable cells within each LLM, and LLMs were weighted equally when averaging across LLMs. Under MICE-PMM, r^* was computed from the coefficients pooled across the five imputed datasets.



Figure 2. Percentage of estimable cells in which each estimated path had the expected direction, by LLM and statistical model.

B4.2 Construct-Level Agreement with Human Ratings (V_h)

Twenty-three raters with backgrounds in organizational behavior and management independently rated 32 randomly selected posts on the same items, following the same instructions as the LLMs. Agreement

among raters was high (mean pairwise r = .675; α across raters = .968). For each of the five constructs, we correlated an LLM's construct scores with the mean human ratings. The five correlations were transformed with Fisher's z, averaged, and transformed back to yield V_h.

B4.3 Measurement Quality

For each LLM, we computed Cronbach's α for the four multi-item scales and fitted a four-factor confirmatory factor analysis (CFA) to the 19 item ratings before imputation, comparing its fit with a one-factor model. Remote work intensity was excluded because it is a single item.

B4.4 Additional LLM Properties

- ***Stability:*** the standard deviation of r^* across an LLM's estimable cells; lower values mean conclusions depend less on the evidence requirements.
- ***Evidentiary strictness:*** the share of items an LLM left missing.
- ***Typology:*** each LLM's position on the two levels of evaluation (Table 11).

Table 11

Typology of LLMs by Construct-Level and Relationship-Level Evaluation

	High r^*	**Low r^***
High V_h	Suitable for theory testing	Accurate scores but distorted relationships (e.g., through missingness or selection of posts)
Low V_h	Most concerning: theory-consistent relationships from inaccurate scores, possibly supplied by the model's prior knowledge	Unsuitable

Note. "High" and "low" are defined relative to the medians across the 12 LLMs.

Table 12 reports these criteria for each LLM. Table 13 and Figure 3 show how model size, agreement with human ratings, and internal consistency relate to similarity r^* across the 12 LLMs.

Table 12

Evaluation Criteria by LLM

Model ID	**LLM (developer)**	**V_h**	**Mean α**	**r^* SEM**	**r^* Path analysis A**	**r^* Path analysis B**	***SD* of r^***	**Missing rate**	**V_h / r^***
1	GPT-5 (OpenAI)	.791	.982	.844	.814	.917	.038	.881	High / High
2	GPT-5-mini (OpenAI)	.736	.988	.671	.658	.932	.033	.692	High / Low
3	Qwen3.8-Flash (Alibaba)	.773	.987	.647	.698	.859	.132	.889	High / High
4	Qwen-Plus (Alibaba)	.613	.974	.451	.505	.664	.164	.805	Low / Low
5	Qwen-Max (Alibaba)	.697	.994	.179	.658	.831	.164	.786	High / Low
6	Qwen3-32B (Alibaba)	.481	.989	.840	.889	.955	.043	.719	Low / High
7	Llama-3.3-70B (Meta)	.618	.988	.678	.709	.954	.041	.561	Low / High
8	Phi-4 (Microsoft)	.391	.945	.412	.770	.844	.100	.564	Low / High
9	DeepSeek-V4-Pro (DeepSeek)	.715	.993	.585	.726	.872	.083	.882	High / High

Model ID	LLM (developer)	V_h	Mean α	*r** SEM	*r** Path analysis A	*r** Path analysis B	*SD* of *r**	Missing rate	V_h / *r**
10	DeepSeek-V4.1-Flash (DeepSeek)	.876	.995	.020	.478	.541	.397	.965	High / Low
11	DeepSeek-R1-Qwen3-8B (DeepSeek)	.146	.978	−.152	−.020	−.022	.078	.638	Low / Low
12	Granite-3.3-8B (IBM)	.015	.760	−.029	.090	.036	.073	.642	Low / Low
-	Median across LLMs	.658	.987	.518	.678	.851	.081	.752	—
-	Correlation with V_h	—	.725	.516	.676	.740	.361	.688	—

Note. V_h = construct-level agreement with human ratings. *r** is averaged across cells with at least 30 posts and all seven path estimates under mean imputation. *SD* of *r** = variation across cells (Path analysis A). Missing rate = share of items the LLM left missing. The last column gives each LLM's position in Table 11. Correlations are computed across the 12 LLMs and are descriptive.

Table 13

Correlations Among Model Size, Agreement with Human Ratings, Reliability, and Similarity

Predictor	Outcome	*N*	Pearson *r*	*p*
Panel A. All 12 LLMs				
Log model size	Agreement with human ratings (V_h)	12	.880	< .001
Log model size	Latent SEM *r**	12	.344	.274
Log model size	Path analysis A *r**	12	.486	.109
Log model size	Path analysis B *r**	12	.524	.081
Agreement with human ratings (V_h)	Latent SEM *r**	12	.516	.086
Agreement with human ratings (V_h)	Path analysis A *r**	12	.676	.016
Agreement with human ratings (V_h)	Path analysis B *r**	12	.740	.006
Mean Cronbach's α	Latent SEM *r**	12	.418	.176
Mean Cronbach's α	Path analysis A *r**	12	.541	.069
Mean Cronbach's α	Path analysis B *r**	12	.606	.037
Panel B. Sensitivity analysis excluding the two 8B models				
Agreement with human ratings (V_h)	Latent SEM *r**	10	−.227	.528
Agreement with human ratings (V_h)	Path analysis A *r**	10	−.475	.166
Agreement with human ratings (V_h)	Path analysis B *r**	10	−.346	.327

Note. Model size was log-transformed before analysis. Panel A includes all 12 LLMs. Panel B excludes DeepSeek-R1-Qwen3-8B and Granite-3.3-8B. V_h = agreement with human ratings; *r** = similarity between the estimated and expected relationship patterns. Path analysis A is the adjusted composite-score path analysis; Path analysis B is the pairwise composite-score path analysis.

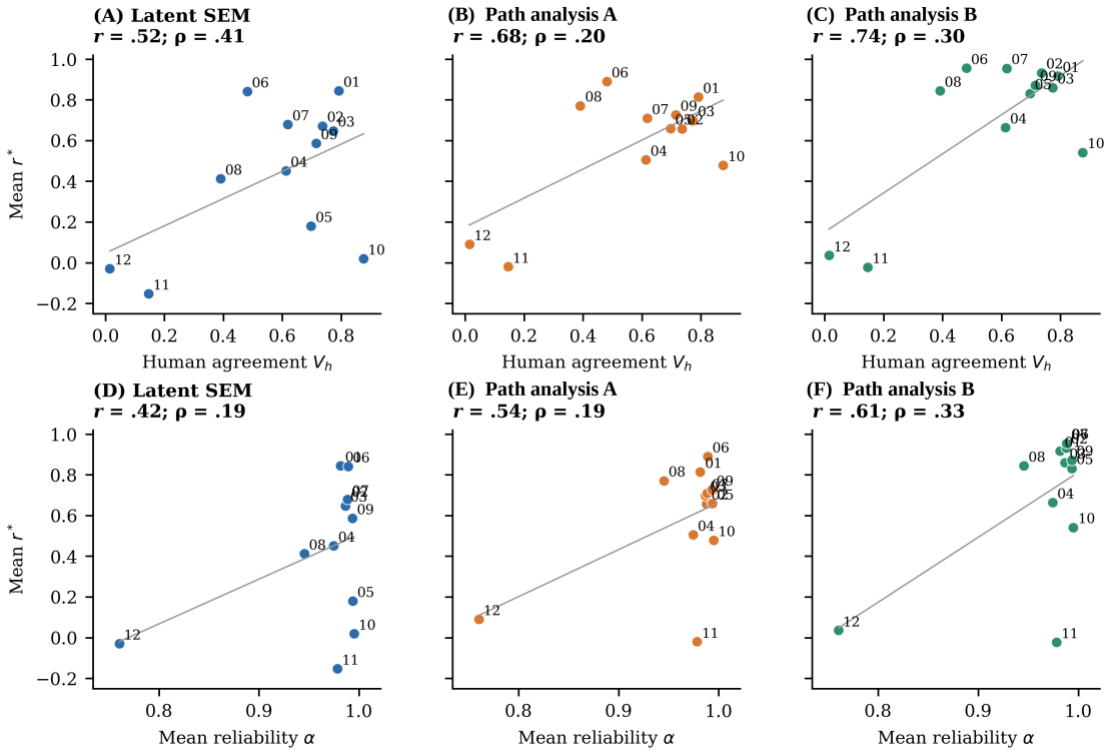


Figure 3. Agreement with human ratings (V_h), internal consistency (mean α), and mean similarity (r*) across the 12 LLMs using the five-construct threshold grid. Lines are descriptive ordinary least squares (OLS) fits. Panel titles give the Pearson (r) and Spearman (ρ) correlations.

B4.5 Locating the Best-Performing Evidence Requirements

To summarize where on the grid relationships came closest to meta-analytic expectations, we recorded, for each LLM and statistical model, the RID–CCB combination with the highest r* among cells retaining at least 30 posts and all seven path estimates. We then plotted the distributions of these 12 RID values and 12 CCB values separately (Gaussian kernel density, bandwidth = .15) and marked the peak of each distribution ± .10 (Figure 4). These distributions summarize where each model most closely reproduced the expected relationship pattern.

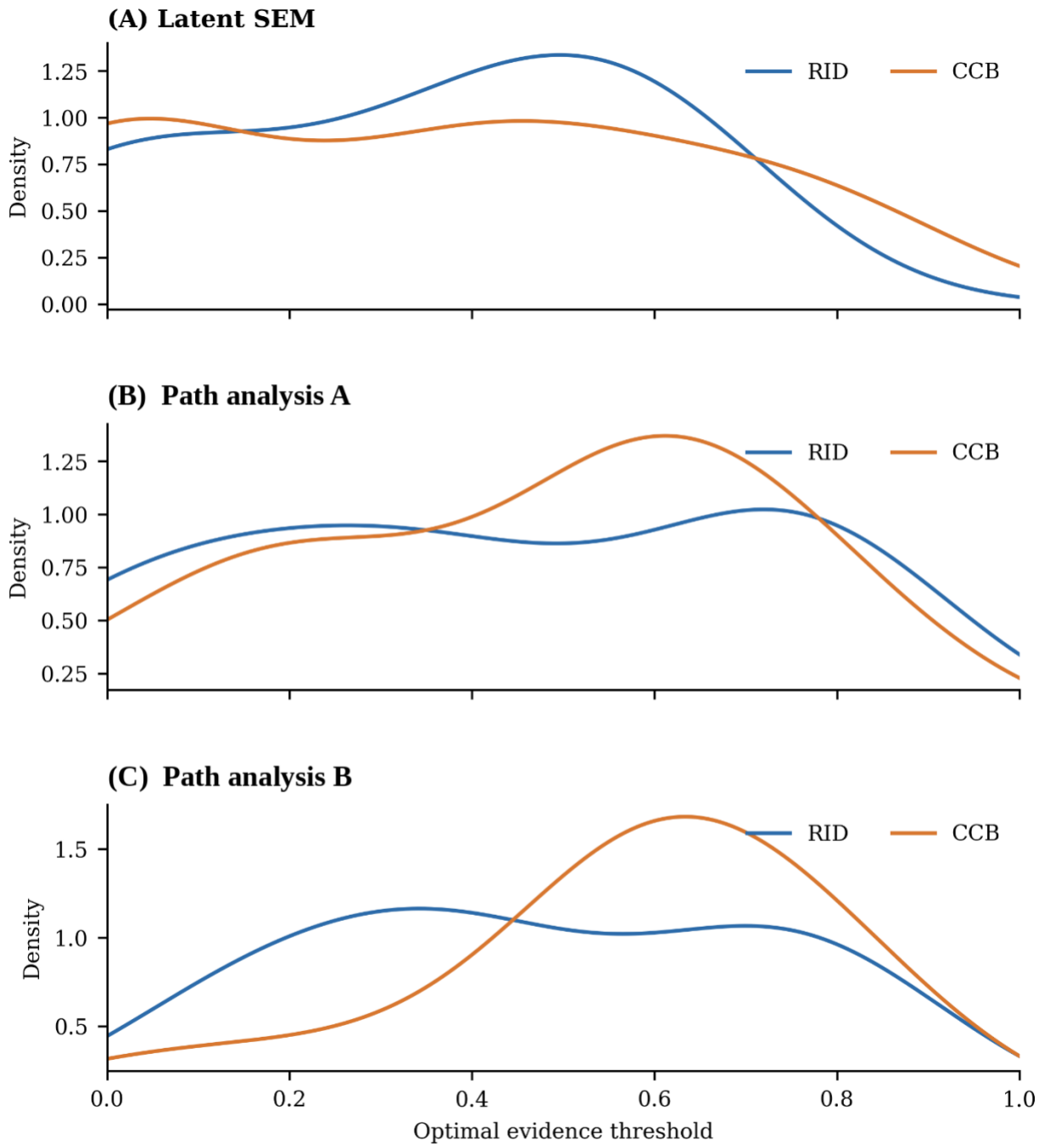


Figure 4. Distribution of each LLM's best-performing five-construct RID and CCB requirements for latent SEM, Path analysis A, and Path analysis B (Gaussian kernel density, n = 12 LLMs).

C. Robustness Check and Planned Analyses

C1. Sample-Size Sensitivity

To test whether results at the median optimal thresholds depend on sample size, we drew random samples of 50 to 5,000 posts at these thresholds (latent SEM: RID = .40, CCB = .40; Path analysis A and Path

analysis B: RID = .40, CCB = .60), 20 times per size for each of the 12 LLMs, and estimated the model under mean imputation. Path analysis A and Path analysis B could be estimated in about 80% of replications at 2,000 posts and 90% at 5,000, whereas latent SEM reached only about 40% at 5,000 (Figure 5). Among estimable replications, mean *r** changed little beyond about 2,000 posts for all three analyses.

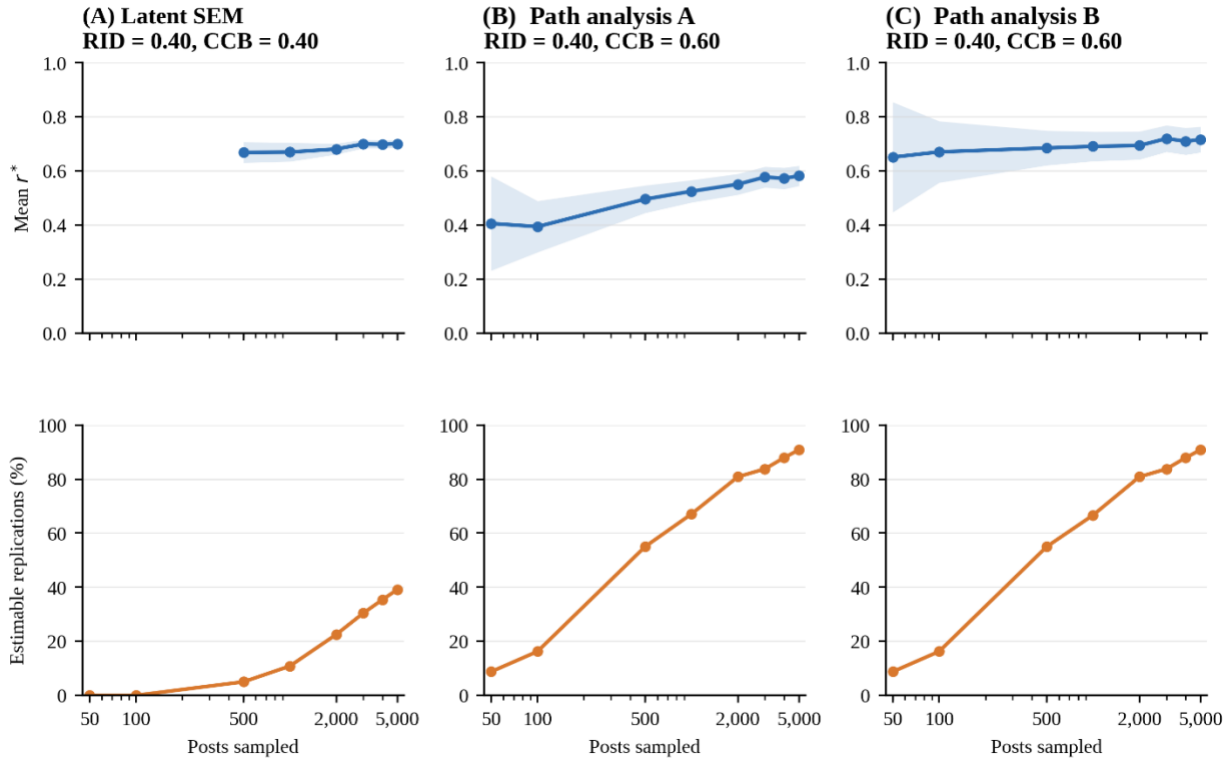


Figure 5. Sample-size sensitivity at representative evidence thresholds. The upper row shows mean *r** among estimable replications; shaded bands show 95% intervals. The lower row shows the percentage of 240 model-by-repetition combinations that could be estimated. Each sample size was repeated 20 times across 12 LLMs under mean imputation.

C2. Analyses Planned for the Development Period

Table 14 lists the analyses we will conduct from January to May 2027, organized by research question.

Table 14

Planned Analyses

RQ	Analysis	Purpose and method
RQ1	Map of acceptable requirements	Express requirements also as the share of posts retained, so they are comparable across LLMs. For each LLM, a cell is acceptable if *r** is within .05 of the LLM's maximum (or within its bootstrap confidence interval) and at least 500 posts remain. Cells acceptable for at least 75% of LLMs form the recommended region. Alternative: the cell that maximizes the lower bootstrap bound of *r**, which penalizes small samples.
RQ1	Variance decomposition	Partition cell-level *r** into variance due to LLM, RID requirement, CCB requirement, and their interactions, separately for each statistical model.
RQ1	Evidence versus sample size	Compare each filtered sample with random samples of the same size, to test whether evidence requirements matter beyond the number of posts retained.

RQ	Analysis	Purpose and method
RQ2	Additional missing-data methods	Compare SoftImpute (Mazumder et al. 2010), missForest (Stekhoven and Bühlmann 2012), and disclosure-aware multiple imputation (DA-MI), which uses both observed ratings and the pattern of topics each post discusses, with the mean-imputation baseline in every cell, focusing on low-CCB posts.
RQ3	Stratified human ratings	Expand human ratings, sampling posts across levels of RID and CCB, across constructs, and where LLMs disagree.
RQ3	Low-evidence and masking tests	Compare r^* in low- versus high-evidence posts, and remove the evidence for a construct from posts to see whether the LLM still produces it. High r^* without evidence would indicate relationships supplied by the model's prior knowledge.
RQ3	Model size × task complexity	Test several sizes within one model family on tasks ranging from explicitly stated to inferred constructs and from single relationships to full models, to identify the smallest sufficient model.
RQ3	Correlates of relationship-level validity	Across LLMs, relate r^* and V_h to evidentiary strictness, median RID and CCB, model size, number of estimable cells, and stability (descriptive; $n = 12$).
RQ3	Cross-model consensus	Test whether an LLM's agreement with the other LLMs predicts V_h, so that researchers without human ratings can screen LLMs.
RQ3	Halo check	For LLMs with very high α, test whether items within a construct receive near-identical ratings, which would indicate impression-based rather than item-specific rating.
All	Generalization and release	Replicate the workflow in a second domain; compare estimates with meta-analytic effect sizes rather than directions alone; release an installable package.

D. Reproducibility and Compute

The anonymized repository [https://anonymous.4open.science/r/research-49E9/] contains versioned task definitions and prompts, the evidence-annotation schema, the rating pipeline, independent analysis modules, a machine-readable archive of all results, fixed-seed scripts that reproduce every figure and table, and a synthetic dataset that runs end to end on open-weight models. Table 15 lists the software, hardware, and data-sharing details.

Table 15

Reproducibility Details

Component	Details
Software	Python 3.12.7; R 4.4.3; lavaan 0.7-2; project implementation of MICE-PMM
Hardware	Local GPUs for open-weight models; CPUs for statistical analysis
Decoding	Greedy decoding for open-weight models; archived request parameters for API models
Random seeds	20261016 for subsampling
Data shared	[Post IDs]; all LLM annotations and ratings; synthetic dataset

E. Prompts

All LLMs received the same prompts. Model-specific wrappers were used only to submit the prompt and read the response.

E1. Evidence Prompt (Step 1)

```
Given one post and a list of items, return its semantic clusters once.

Definitions:
- An occurrence is one independently removable meaning expressed in the post. Split independent
claims, but do not split stylistic elaboration.
- Equivalent occurrences express the same meaning in context. Keep repetitions as separate
occurrences and give them the same cluster_id. Related but different meanings need different
clusters.
- substantive is false only for greetings, boilerplate, navigation text, or content-free filler.
Facts, experiences, evaluations, emotions, intentions, preferences, and concrete questions are
substantive.
- Form clusters solely from meanings expressed in the post, before considering the items. Items
must never create, name, split, or merge clusters.
- Each cluster contains exactly one subject-predicate claim. Never join independently true claims
with "and", "but", a comma, or a semicolon; output separate clusters.
- occurrence_count counts only contextual paraphrases or repetitions of that same proposition,
never the number of facts inside a sentence.
- A cluster is relevant to an item only when that atomic proposition alone can serve as direct
evidence in an answer to the full item text. Do not propagate relevance from nearby clusters or
the post as a whole.
- Direct evidence must answer the item's main interrogative predicate, not merely repeat a
keyword. Use the full scope, not just the item title.
- When an item enumerates requested variables or categories, facts outside that enumeration do
not match unless the item explicitly includes a catch-all category.
- Shared topic, setting, actor, neighboring negative affect, possible cause, possible
consequence, or an unstated construct is not direct evidence.
- After forming the clusters, compare every cluster with every item. Include all and only item
IDs for which the cluster is direct answer evidence; do not leave relevant_items empty when
direct evidence exists.

Requirements:
1. Cover all substantive meanings in the post, including background facts that match no item.
2. Put equivalent repetitions in the same cluster and report their occurrence_count.
3. Do not force a cluster to match an item; relevant_items may be empty.
4. Keep each meaning at no more than 18 words. Do not quote or explain the post.
5. Return compact JSON only and stop after the JSON.
6. Every clusters element must be an object; do not use cluster IDs as JSON keys or duplicate
fields.
7. Quote every item ID, use an integer occurrence_count, and do not emit backslash escapes in
meaning.
8. occurrence_count must be a positive integer of at least 1, including for substantive=false
clusters.
9. Never return an empty clusters list. If the post contains only filler, represent each filler
meaning with substantive=false and occurrence_count at least 1.

{
  "clusters": [
    {
      "cluster_id": "C1",
      "meaning": "one atomic proposition",
      "substantive": true,
      "occurrence_count": 1,
      "relevant items": ["I1"]
    }
  ]
}

Items:
{items}

Post:
<post>
{post}
</post>
```

E2. Rating Prompt (Step 2)

```
You are scoring one Reddit post with the same constructs used by trained human raters.

Identify every distinct chronological or counterfactual work stage described in the post. If
there is no meaningful change, return one stage. Score only information supported by the supplied
post. Do not infer a score from stereotypes. Use null when the construct cannot be scored from
the text.

Exactly one stage must have "selected_for_analysis": true. Select the stage identified by the
study's focal-stage rule supplied below. If no separate rule is supplied, choose the
primary/current work situation that is most directly described and has the strongest joint
evidence for remote-work intensity, autonomy, stress, satisfaction, and LMX. Other stages must be
false.

Construct item scores use integers from 1 to 7. Remote work intensity uses 0 to 7.

Remote work intensity:
0 = fully on-site/no remote work, 7 = fully remote. Intermediate values represent increasing
remote-work frequency.

Perceived job autonomy items:
1. Independence from supervisor in deciding place of work.
2. Independence from other people in defining work location.
3. Discretion in deciding where to work.
4. Discretion in defining work schedule.
5. Independence from supervisor in defining work schedule.
6. Independence from others in defining work schedule.

Work-related stress items:
1. Workload feels heavy.
2. Difficulty meeting conflicting demands.
3. Generally feels stressed by the job.

Job satisfaction items:
1. All in all, satisfied with the job.
2. Likes the job (score the reversed original wording so higher always means greater
satisfaction).
3. Likes working there.

LMX items:
1. I know where I stand with my leader and how satisfied they are with what I do.
2. My leader understands my job problems and needs.
3. My leader recognizes my potential.
4. My leader would use their power to help me solve problems in my work.
5. My leader would bail me out at their expense.
6. I have enough confidence in my leader that I would defend and justify their decisions if they
were not present.
7. I have a high-quality working relationship with my leader.

Return one JSON object only:
{
  "stages": [
    {
      "stage_index": 1,
      "stage_label": "short evidence-based label",
      "selected_for_analysis": true,
      "remote_work_intensity": 1,
      "autonomy_items": [1, 2, null, null, null, null],
      "stress_items": [1, 2, 3],
      "satisfaction_items": [1, 2, 3],
      "lmx_items": [1, 2, 3, 4, 5, 6, 7]
    }
  ]
}

STRICT OUTPUT REQUIREMENTS:
```

```
- Output JSON only. Do not add Markdown, analysis, explanation, or text after the closing brace.
- Each array contains item SCORES IN THE FIXED QUESTION ORDER, not item numbers.
- autonomy_items must always contain exactly 6 positions.
- stress items must always contain exactly 3 positions.
- satisfaction_items must always contain exactly 3 positions.
- lmx_items must always contain exactly 7 positions.
- Preserve every position even when unsupported: use null in that position. Never shorten an
array.
- Valid construct scores are integers 1 through 7 or null. Never use 0 for a construct item.
- remote_work_intensity alone may use integers 0 through 7 or null.
- Before responding, count the four arrays: 6, 3, 3, and 7.

POST:
{post}
```

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